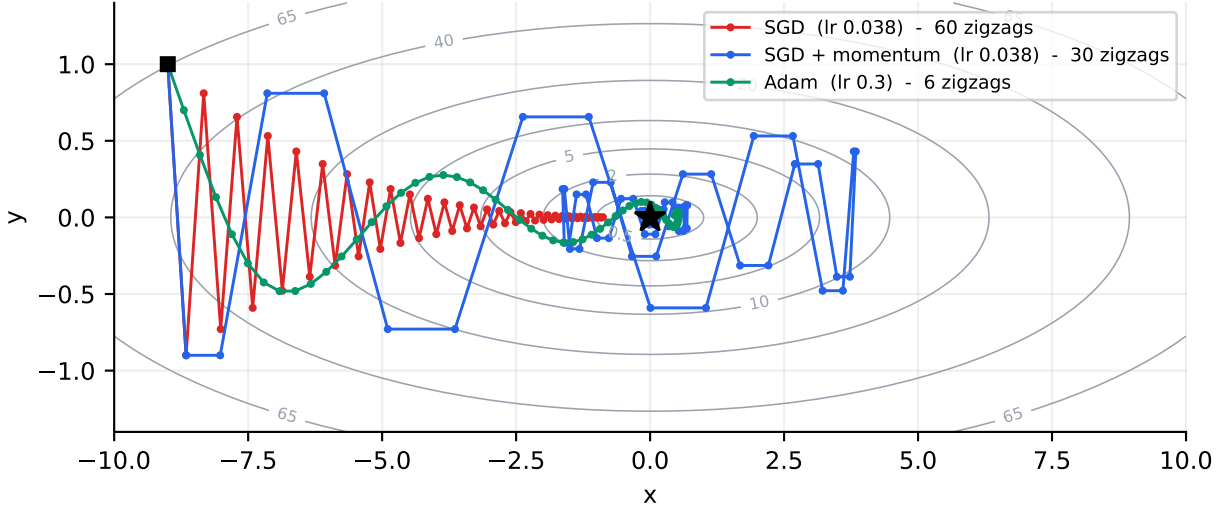
Narrow valley: $f(x, y) = \frac{1}{2}(x^2 + 50y^2)$ - start at $(-9, 1)$, 60 steps
SGD (lr 0.038) - 60 zigzags
SGD + momentum (lr 0.038) - 30 zigzags
Adam (lr 0.3) - 6 zigzags
65
40
5
2
10
65
1.0
0.5
0.0
−0.5
−1.0
y
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x